

Beyond the Rules: Understanding the Design Logic of Internet Standards

SemTech 2026

Jie Bian, Michael Welzl, Nikolay Arefyev

June 29, 2026

University of Oslo

Background: RFC, IETF & Email Archives

RFC (Request for Comments)

What it is:

Formal technical documents defining internet protocols & standards

Key facts:

Numbered sequentially (e.g., RFC 793 = TCP, RFC 8200 = IPv6)
Published by IETF after community review & rough consensus
Over 9,000 RFCs published since 1969

IETF (Internet Engineering Task Force)

What it is:

Open standards body developing internet protocols since 1986

How it works:

Organized into Working Groups (WGs) by topic area
Decisions made via rough consensus in public discussion
Open to all participants

Output:

Publishes RFCs as the final product of standardization

IETF Email Archives

What it is:

Public mailing lists where WGs debate design choices

Access:

Fully public and searchable at <https://mailarchive.ietf.org/arch/>

Motivation: The WHY is Missing

- RFCs specify WHAT to implement — not WHY a design was chosen
 - Rationale is buried in IETF email archives: more than 20 years, ~130,000 emails/year, more than two million emails, exchanged among 56 000 participants, over 500 mailing lists
- Example: RFC 6817 sets INIT_CWND = 2. Why not 4?
 - The answer exists in email discussions — but it takes hours to find, or may never be found at all
 - LLMs (Gemini 3, Claude Opus 4.5) need several rounds of conversation to point to a relevant email thread
- Who suffers from this gap?
 - Newcomers: lack essential context for meaningful participation in the standards process
 - Implementers: cannot align with a specification's intent without understanding the reasoning behind it
 - Future protocol designers: cannot understand the trade-offs that shaped earlier standards, leading to repetition of past mistakes

Goal: Automatically Retrieve and Generate Design Rationales



- **We are the first to introduce an automated methodology — grounded in RAG — that identifies, retrieves, and synthesizes design-decision rationales from the extensive IETF email corpus.**
 - Our system retrieves rationale-rich context from extensive IETF email archives and WG deliberations, then generates clear explanations of the motivations and constraints behind protocol design decisions.

Dataset: Annotated Decision–Email Pairs

Three RFCs with manually annotated decision–rationale (email or summarized explanation) pairs

RFC	Pairs	Notes	Role
RFC 7657	163	DART WG — delay-based DSCP marking	Retrieval
RFC 8335	72	INTAREA WG — PROBE message for ICMP	Retrieval
RFC 8205/8374	46	SIDR WG — BGPsec; gold rationales available	RAG

RFC 8374 serves as a retrospective summary of the development of RFC 8205 (the BGPsec protocol), written by an author directly involved in RFC 8205’s drafting and therefore providing an explicit account of key decisions and their underlying discussions.

- **Two retrieval scopes evaluated**

- Draft-level: only emails matching the specific I-D name— easier
- WG-level: all emails from the entire working group— harder, captures discussion threads that draft-level filtering misses

Synthetic Data Generation: Bridging the Domain Gap

- **Problem:** no labeled email $\leftrightarrow$ RFC-decision pairs exist for training
- **Solution:** use LLMs to synthesize RFC-style decision statements from emails
 - Models: Qwen2.5-32B and Llama-3.3-70B — two model families for diversity
 - Pre-filter: model first judges whether an email is technical before generating
- **Hard positive strategy**
 - Email threads accumulate quoted replies $\rightarrow$ emails share lots of surface text
 - Non-verbose version: remove quoted content before training
 - Forces the retriever to go beyond topical similarity — the relevant rationale may not look like the decision text at all, so the model must learn what is **causally relevant**
- **Scale:** ~350K decision-email pairs per model family (Qwen and Llama)

Retrieval: Fine-Tuned Dense Retrievers vs. Baselines

- **Sparse Baseline**
 - BM25
- **Dense Baseline**
 - Promptriever — instruction-tuned dense model (Mistral-7B) for general retrieval tasks
- **In-domain Baseline**
 - M(R) — trained on real-world IETF GitHub decision–rationale data (prior work)
- **Our models (Mistral-7B backbone)**
 - M(LN) M(QN)— fine-tuned on Llama or Qwen synthetic email data, non-verbose version
- **Training setup**
 - Contrastive loss with in-batch negatives
 - PEFT fine-tuning — only ~1% of parameters updated
 - Batch: 24 across 2 GPUs; 1 positive + 47 negatives per instance
 - Verbose email truncated to 4096 tokens; non-verbose truncated to 3584; queries to 512
- **Evaluation metrics**
 - Hit@3, Hit@10 — is target in top-K?
 - MRR@1000 — how highly is the target ranked?

Retrieval Results: Synthetic Training Closes the Domain Gap

- **Finding 1:** On MRR, BM25 and M(QN) perform comparably across both RFCs and retrieval scopes — with only marginal differences between them.
- **Finding 2:** On HIT@3, M(QN) consistently outperforms BM25 at both scopes — the advantage is clearer at WG-level, e.g., RFC7657 (DL→WG-L): M(QN) 66.67% → 58.97% vs BM25 61.54% → 48.72%
- **Finding 3:** Promptriever outperforms M(R) but falls below BM25 — dense retrieval alone, without domain adaptation, is insufficient.
- **Finding 4:** M(R) — trained on real GitHub data — transfers poorly to email domain
 - GitHub comments are concise, structured, and task-focused; emails are longer, informal, with context setting, digressions, and social interactions
- **Finding 5:** Non-verbose (de-quoted) training consistently beats verbose
 - Removing quoted content creates cleaner, more discriminative training signal

RAG: Generating Rationales from Retrieved Emails

- **Generation**

- Generator: Llama-3.3-70B
- Prompt style inspired by RFC 8374 rationale text
- Produces concise, evidence-grounded rationale
- No new model training — zero-shot prompting

- **Challenge**

- Emails are long, informal, multi-viewpoint
- No single email contains a clean, authoritative rationale
- LLM must distill and synthesize across retrieved context

- **Automated (LLM-based) Evaluation**

- Context Recall: how much of the reference rationale is covered by retrieved emails
- Faithfulness: does the generated text stay grounded in the retrieved context?
- LLM-as-Judge: Llama-3.3-70B rates overall rationale quality

- **Human as Judge**

- A domain specialist rates generated rationales on a 1–4 scale, blinded to retrieval method

- **Test case** (RFC 8205 / RFC 8374)

- 37 decisions retained — 9 removed as they contained no actionable design justification or final recommendation

RAG Results: Comparable Quality at 10× Lower Token Cost

Retriever	Avg email (chars)	Context Recall	Faithfulness	Response (chars)	Expert Rate
BM25	21,955	70.49	68.67	598	2.27
M(LN)	2,038	49.62	53.74	450	2.30
M(QN)	3,409	55.72	59.06	486	—

- BM25 retrieves quote-heavy emails → 10× longer context, higher lexical recall
- Dense retrievers select focused emails → comparable expert rating with far fewer tokens
- Human evaluation: expert found LLM explanations 'consistently useful'
 - Only 2–3 cases not fully self-contained; 1 minor factual inaccuracy observed
 - Style diverges from gold reference — but substance is rated comparably
- Conclusion: dense retrieval matches human-judged quality at 10× lower token cost — RAGAS metrics favour BM25 due to its longer retrieved context

Conclusion

- **Contributions**

- First automated RAG system for RFC design-decision rationale extraction
- Synthetic data generation pipeline that transfers to the email domain
- Non-verbose hard positive strategy improves retrieval
- Dense retrieval achieves comparable generation quality at 10× lower token cost

- **Key takeaway**

- Making protocol rationale accessible is feasible — and useful to real experts

- **Limitations**

- Evaluation covers only 3 RFCs — limited generalizability
- Gold reference rationales available for only RFC 8205
- Generated style diverges from author-written text

- **Future work**

- Explore methods that combine the strengths of sparse and dense retrieval
- Larger annotation effort — collecting more expert-written rationales across more RFCs
- Document-side representation refinement: thread condensation, discourse unit extraction, or salience-based filtering